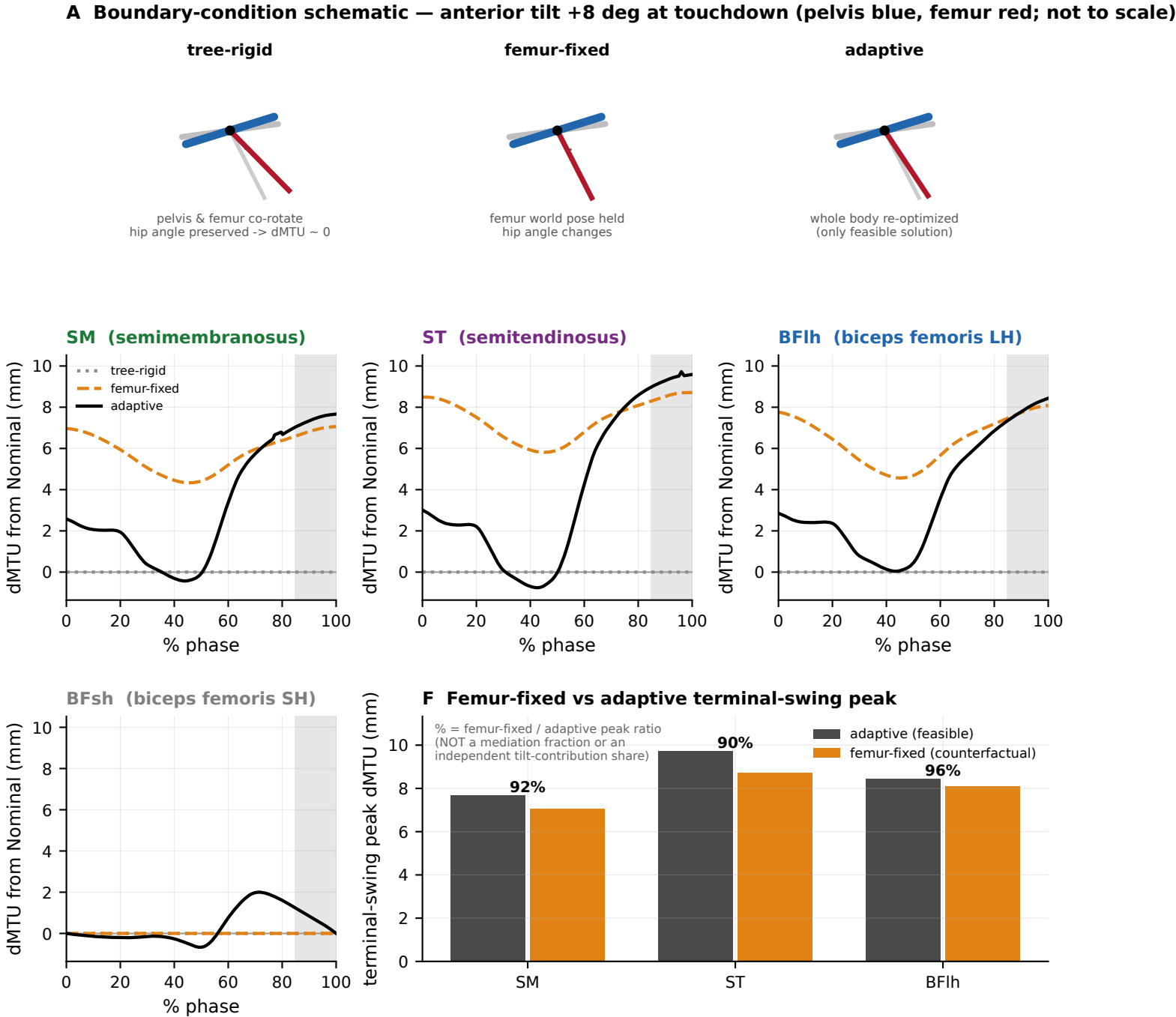


Figure 5 | Pelvis-femur coordination decomposition of biarticular hamstring MTU stretch



Terminal swing = last 15% of phase (boundary-analysis window). Tree-rigid & femur-fixed are geometric counterfactuals (not runnable); adaptive is the re-optimized N=100 max-tilt solution. MTU length from OpenSim 4.4 exact geometry, phase-normalized.